Experiment 6

Title: Implementation of Chunking to Extract Noun Phrases

Aim

To implement **chunking techniques** for extracting noun phrases from a given text.

Objectives

1. To understand the concept of chunking in Natural Language Processing.
2. To study noun phrase extraction using chunking methods.
3. To implement chunking for identifying noun phrases in a sentence.

Theory

1. Introduction to Chunking
Chunking is a technique in **Natural Language Processing (NLP)** used to group words into meaningful units called chunks. These chunks represent parts of a sentence such as noun phrases or verb phrases. Chunking helps in understanding the grammatical structure of a sentence.

2. Noun Phrases
A noun phrase is a group of words that **acts as a noun in a sentence**. It usually consists of a noun along with its modifiers such as adjectives, determiners, or articles. For example, in the phrase *"the big red car"*, the entire group of words forms a noun phrase.

3. Chunking for Noun Phrase Extraction
Chunking identifies and extracts noun phrases by using **grammatical patterns and POS tags**. After performing POS tagging, rules are applied to detect patterns like *determiner + adjective + noun*. This helps in automatically extracting meaningful noun phrases from text.

4. Applications of Chunking
Chunking is widely used in **information extraction, question answering systems, text summarization, and search engines**. It helps computers identify important entities and understand the structure of sentences more effectively.

Conclusion

In this experiment, chunking was implemented to extract noun phrases from text. The method groups related words together based on grammatical patterns. This helps in identifying

meaningful units within a sentence. Noun phrase extraction improves the understanding of sentence structure in NLP tasks. Therefore, chunking is an important technique in text processing and language analysis.